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Innovation and entrepreneurship education for medical students: a global bibliometric analysis (2000–2024)

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ABSTRACT

College students are important contributors to global innovation and entrepreneurship, making it increasingly important to improve related education, especially in medical schools. However, challenges remain due to the specific nature of medical training, economic differences across countries, and the lack of a well-developed system. In some regions, limited understanding has led to ineffective efforts and poor results, which has hindered progress, underscoring the need for a strategic, context-aware approach to innovation and entrepreneurship education in medical schools. This study employs bibliometric methods including Microsoft Excel, CiteSpace, VOSviewer and R language package of Bibliometricx software to analyze global publications published on Web of Science database from 2000 to 4 December 2024.The findings indicate that innovation and entrepreneurship education for medical students began later than in other fields and has faced implementation challenges. Publication trends align with national policies and significant events, such as the COVID-19 pandemic. Developed countries dominate this field, while international collaboration has exacerbated regional disparities. Higher medical schools remain the primary contributors, reflecting the limited scope of this research area. Current studies emphasize the learning aspects of medical education but insufficiently address students' innovation and entrepreneurship abilities. The results also highlight critical gaps in current education models and suggest that integrating innovation and entrepreneurship more comprehensively into medical curricula is crucial for preparing students for the evolving healthcare landscape.Medical education must adopt an interdisciplinary approach, as global public health developments have shaped its trajectory. This study informs health policy by showing that innovation and entrepreneurship education strengthens medical students' ability to address global health challenges. It provides guidance for curriculum integration through interdisciplinary and context-driven approaches, underscoring the need to cultivate a culture of innovation to advance medical education and global health outcomes.

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Medical student; innovation education; entrepreneurship education; bibliometric; medical education

Introduction

College students represent a group with remarkable innovative enthusiasm and creative potential. They also serve as a vital fresh and reserve force for advancing innovation-driven development strategies and fostering innovation and entrepreneurship on a global scale. Strengthening innovation and entrepreneurship education in colleges and universities is a crucial initiative to enhance the integration of higher education with economic and social development. It also constitutes an intrinsic requirement for achieving sustainable economic and social progress [1]. As an essential component of higher education, the reform and advancement of medical education in the context of innovation and entrepreneurship highlight its distinctive characteristics and necessity, particularly under the significant influence of the scientific and technological revolution and the rapid pace of globalization [2,3].

However, the distinctive characteristics of medical education, including higher entry requirements, diverse disciplines, rigid theoretical frameworks, prolonged study duration, a strong focus on memorization, and reliance on empirical knowledge, contribute to longer implementation cycles, slower progress, higher costs, and less significant outcomes in innovation and entrepreneurship education compared to other science and technology fields [4]. Furthermore, disparities in global economic development have led to substantial differences among countries in the demand for medical professionals, training objectives, educational systems, and investments in training costs.

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Given the increasing recognition of these needs, it is important to examine how innovation and entrepreneurship are currently taught in medical education and to identify the main obstacles limiting their development and impact. Despite over fifty years of efforts to promote innovation and entrepreneurship education for medical students, fundamental issues such as simplistic, experience-based methods continue to persist [5]. Currently, no mature or replicable training system exists, which negatively impacts sustainable development and scientific implementation. As a result, significant challenges persist in ensuring the sustainable growth and effective execution of such education.

Innovation and entrepreneurship education for medical students is a critical issue that plays a vital role in cultivating exceptional medical talents and advancing human health, holding significant importance for promoting global health. With the acceleration of global integration and the ongoing scientific and technological revolution, more countries have prioritized reforms in this area [6–9]. However, many countries and institutions still have a limited understanding of what should be done and how to implement it. This is often due to fixed assumptions or the tendency to follow trends blindly, without fully considering their own contexts and the specific nature of medical education. Such misguided efforts lead to inefficient results, limiting medical students' individuality and creativity, while impacting the quality of education.

The primary obstacle lies in the lack of clarity regarding the developmental history and knowledge mapping of innovation and entrepreneurship education for medical students in most national medical higher education institutions. This situation highlights an urgent need for a scientific and quantifiable research methodology to explore the developmental processes, patterns, hotspots, and trends within this field. Bibliometrics and visual analytics have emerged as valuable and effective approaches to address this critical issue [10–12].

Consequently, this study aims to fill the gap by providing an in-depth analysis of the historical development and current state of innovation and entrepreneurship education in medical schools, identifying patterns and trends that have shaped its evolution. Here, we apply bibliometric methods to analyze global publications from 2000 to 4 December 2024. Our analysis includes the number of publications, citations, journals, national and international collaborations, authors, keyword co-occurrence, and hotspot analysis. The goal is to provide policymakers, administrators, educators, and medical students in medical education institutions worldwide with a comprehensive understanding of the development of innovation and entrepreneurship education for medical students in the 21st century. By doing so, we aim to assist them in gaining deeper insights into, optimizing, learning from, and practicing the most advanced and cutting-edge research in this field.

Methodology and materials

Research method: bibliometric

In recent years, bibliometrics has emerged as a key tool in the analysis of literature reviews, with widespread applications in both academic and industry intelligence fields [13]. Specialized software enables the visualization of large-scale literature data, revealing underlying network relationships and the knowledge structure [14,15]. This approach provides quantitative insights into research trends, offering robust support for the objectivity and scientific integrity of conclusions [16].

For this study, bibliometric information was extracted from the WoSCC database, including authors, titles, journals, publication years, countries/regions, institutions, keywords, citations, abstracts, and references, and saved in plain text format. Impact Factor (IF) data was obtained from the Journal Citation Reports 2023. The analysis tools included Microsoft Office Excel, CiteSpace (6.3.R3), VOSviewer (1.6.20), and the Bibliometricx package based on R language (4.4.2) (4.30). To improve the accuracy and representativeness of the analysis, data preprocessing was carried out in accordance with the specific characteristics of the tools, such as merging different names for the same institution, consolidating secondary institutions under primary ones, and standardizing keywords with similar or identical meanings.

Data source

Web of Science is a widely recognized comprehensive database in the academic community, with the literature it includes undergoing rigorous peer review and publication scrutiny [17,18]. For this study, literature was selected from the Web of Science Core Collection, specifically from the Science Citation Index Expanded (SCIE) and the Social Sciences Citation Index (SSCI), which cover high-quality research in the fields of science and social sciences, ensuring the comprehensiveness and reliability of the data.

Inclusion and exclusion criteria

We conducted topic-based searches by combining the keywords 'medical student', 'innovation', 'entrepreneurship', and 'education', with the following search strategy: TS = (('medical student*') AND ('innovat*' OR 'dual-innovation' OR 'entrepr*' OR 'startup*' OR 'start-up*') AND ('instruct*' OR 'teach*' OR 'education')).The publication date was restricted to between 1 January 2000, and 4 December 2024, with the literature types limited to 'Article' and 'Review' and language set to 'English.' The search was capped at

4 December 2024, to avoid potential data discrepancies from database updates.

Manual screening was conducted to ensure the validity and reliability of the data. To reduce subjectivity, three researchers independently screened the retrieved articles. Discrepancies were resolved through discussion and voting. The screening process followed the PRISMA guidelines [19], encompassing four steps: retrieval, initial screening, inclusion, and synthesis. The initial search yielded 1,394 journal articles. A total of 96 items – such as book reviews, editorials, corrections, news articles, brief reports, conference abstracts, and reviews – were excluded. Additionally, 33 non-English articles were removed. Remaining articles were individually assessed for relevance, ensuring the final selection focused specifically on medical student innovation and entrepreneurship education. Duplicate checking via CiteSpace confirmed no redundant records.

To further reduce bias, standardized inclusion criteria were applied. First, studies were included if the sample consisted of medical students, even if the term did not appear in the title [20]. This included undergraduate medical students [21], medical postgraduates, international medical students, and nursing students. Second, studies were included if they addressed innovation and entrepreneurship education – defined to encompass technological innovation [22], institutional innovation [23], or relevant systematic reviews [24]. Articles not explicitly focused on medical students or on innovation/entrepreneurship education were excluded. Ultimately, 1,265 valid articles from the Web of Science Core Collection were included. Figure 1 presents the study's overall framework.

Bibliometric analysis methods

This study employed multiple bibliometric techniques to identify research trends and developmental directions in the field of medical student innovation and entrepreneurship education. The specific methods are as follows:

Citation Analysis: Used to assess the academic impact of articles, journals, and authors by citation frequency [25], enabling identification of high-impact publications and core literature.

Co-author Analysis: Applied to explore collaboration networks among researcher [26]. VOSviewer was used to visualize co-authorship structures and major collaboration clusters.

Keyword Co-occurrence Analysis: Employed to reveal thematic structures and interdisciplinary linkages based on the co-occurrence of keywords [27]. Both CiteSpace and VOSviewer were used to conduct clustering analysis and identify key themes and trends.

Burst Detection: Used to identify emerging topics that experienced rapid increases in academic attention [28]. CiteSpace detected research hotspots related to COVID-19, including 'online learning' and 'virtual reality.'

Journal Metrics Analysis: Applied to evaluate journal influence through publication volume, citation counts, and impact metrics [29]. Bradford's Law was implemented using the Bibliometrix package in R (version 4.4.2, package 4.30) to identify and assess core journals in the field.

Technical implementation of bibliometric analysis

In this bibliometric study on medical student innovation and entrepreneurship education, we applied defined threshold criteria and software settings to ensure analytical rigor and accuracy.

Microsoft Office Excel 2021 was used to plot annual publication volumes and proportions.

VOSviewer (version 1.6.20) was used to construct and visualize co-authorship and keyword co-occurrence networks among countries/regions and institutions. In the maps, node size reflects publication volume, line thickness indicates collaboration strength, and color represents clustering. For country-level collaboration (Figure 2a,b), a minimum threshold of five publications was applied, resulting in the inclusion of 43 countries/regions.

CiteSpace (version 6.2.R3) was used for keyword clustering and burst detection. All keywords with a frequency ≥ 1 were included. A burst strength > 2 was set to identify emerging hotspots. Parameters included: time span (2000–2024), 1-year time slices, g-index ($k = 25$), LRF = 2.5, LBY = 5, $e = 1.0$. Network quality was assessed using modularity ($Q > 0.3$) and silhouette ($S > 0.7$) scores [30,31]. In this study, $Q = 0.3984$ and $S = 0.7104$ indicated a clear and reliable cluster structure.

R (version 4.4.2) with the bibliometrix package (version 4.3.0) was used for journal and thematic evolution analysis. With time intervals set at 2013, 2018, and 2022, we tracked topic dynamics and research trend shifts over time.

Result

Time trend of the publications

To better illustrate the changes in publication volume within the field of innovation and entrepreneurship education, we analyzed the annual publication volume using a line chart and recorded the cumulative publication volume with a bar chart in Excel, with a dashed line fitted to represent the annual trend (S Figure 1a). The figure divides the development of the field over the past two decades into three

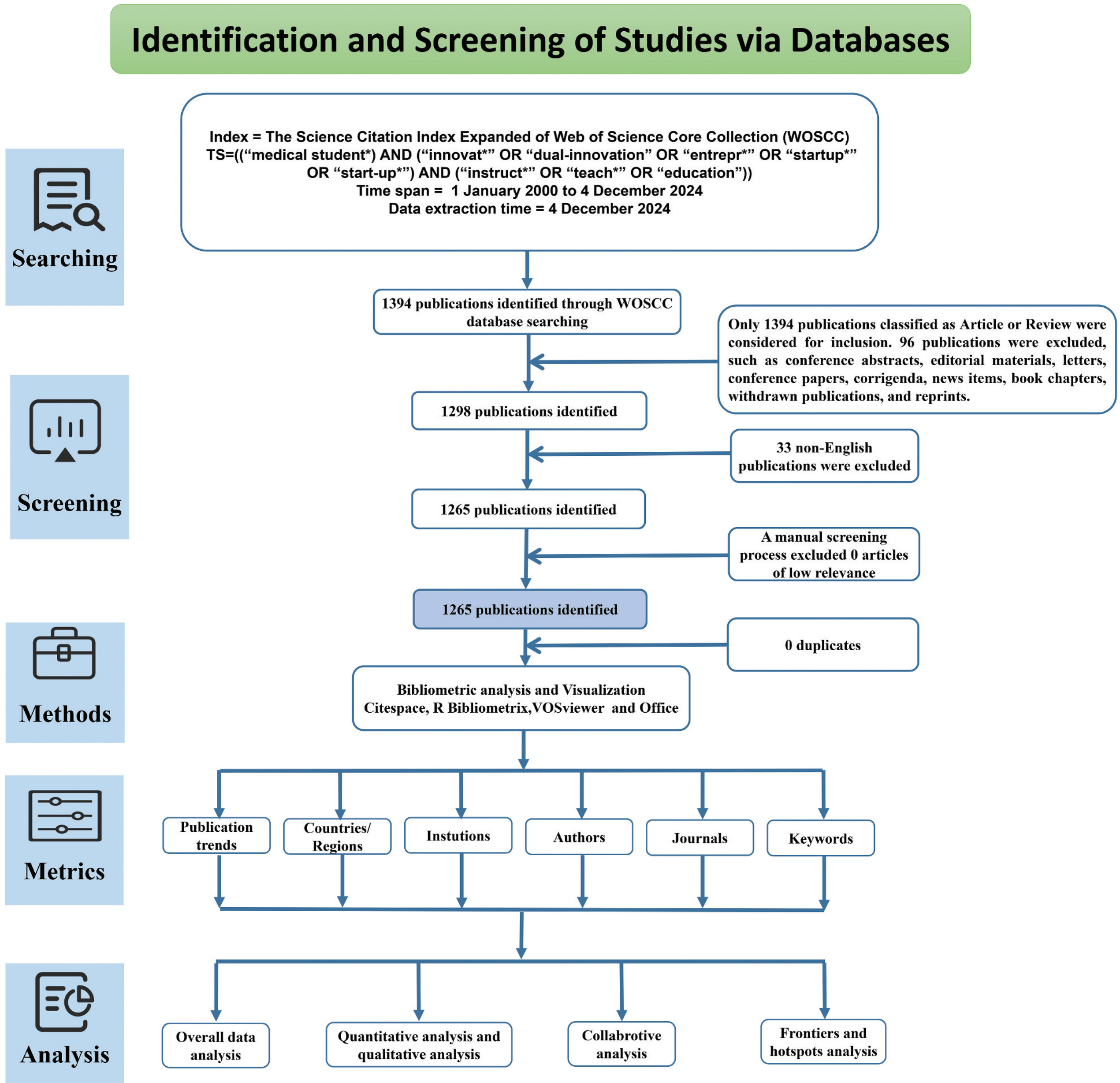

Figure 1. PRISMA flowchart for including studies to review. Limitations on search terms and publication dates yielded 1394 publications. Further restrictions on publication type and language resulted in a final selection of 1265 publications for bibliometric analysis, covering six aspects: publication trends, countries/regions, institutions, authors, research journals and keywords.

stages. The first stage, from 2000 to 2014, saw relatively low publication volumes, influenced by factors such as limited technological maturity and insufficient academic attention to medical students' innovation and entrepreneurship education. During this period, the academic community did not fully recognize the significant research gap in this area or the potential to expand the role of teaching and learning in both spatial and temporal dimensions. The second stage, from 2015 to 2020, marked an increasing focus from scholars. The third stage, from 2021 to 2024, was shaped by the pandemic, which led many scholars to investigate how the shift to online education during this period provided new momentum for medical innovation and entrepreneurship education, offering fresh research directions and new opportunities for medical students. As a result, this area became a key point of focus for both the medical and educational communities [32,33].

To fully understand a research field, it is crucial to first analyze its fundamental quantitative data, with changes in annual publication volume being the clearest indicator of the field's development trend. By examining the temporal distribution and growth patterns of the literature, we can gain insights into the development pace, research level, and scale of the field, while also predicting its future trends and research dynamics [34]. S. Figure 1b provides a detailed overview of the descriptive statistical characteristics of the literature in the field of medical

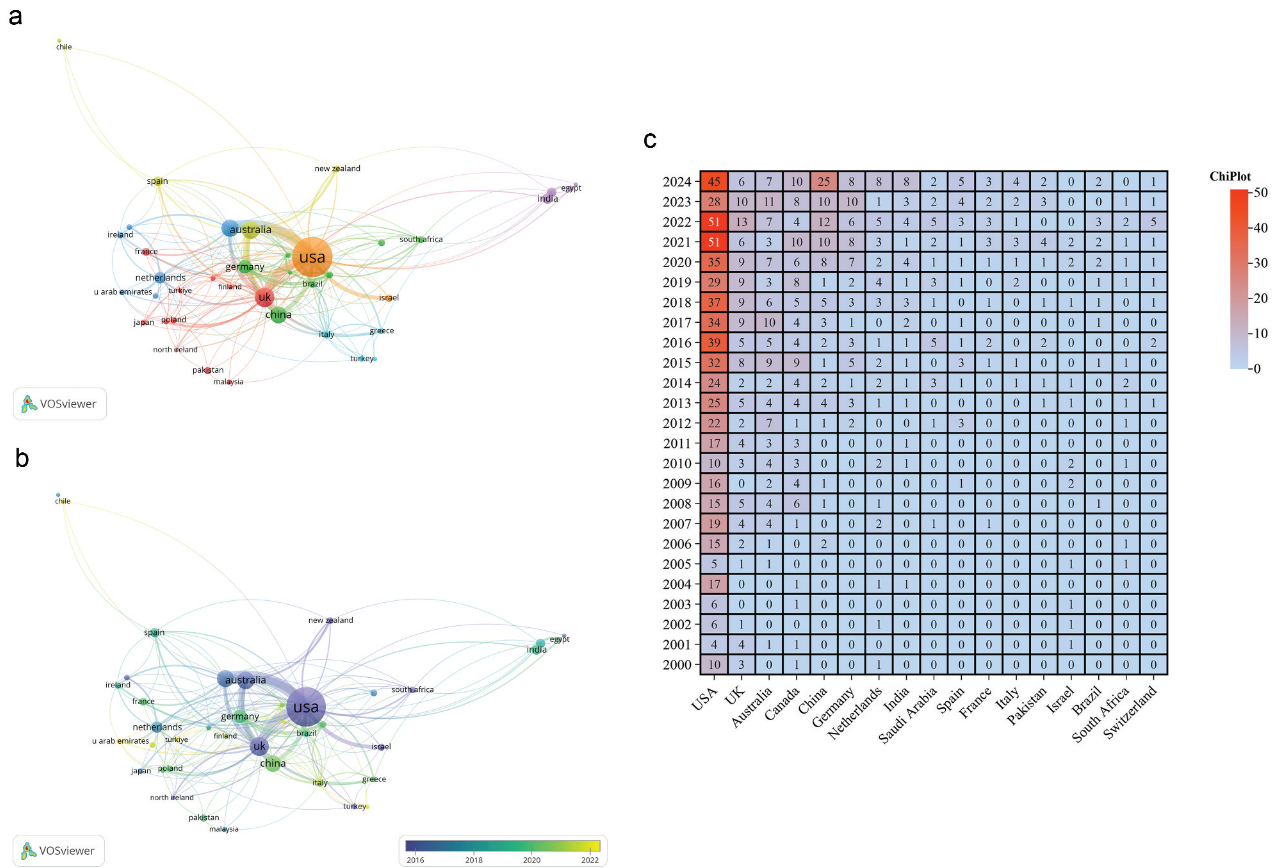

Figure 2. Country co-authorship analysis. The size of each circle indicates the number of articles produced by that country. The distance between any two circles indicates the relatedness of their co-authorship link, and the thickness of the connecting line indicates the strength of the link. A) country co-authorship network visualization map. The color of each circle indicates the cluster it was grouped into by VOSviewer. B) country co-authorship overlay visualization map. The color of each circle indicates the average publication year for the country, according to the color gradient shown in the lower right corner. C) annual fluctuations in academic output for the high-productivity countries/regions (top 15) from 2000 to 2024.

innovation and entrepreneurship education. After deduplication using Citespace software and statistical analysis with R bibliometrics, the final sample size for this study was determined to be 1265 articles, as of the data retrieval cutoff date. These articles were authored by 6231 researchers, representing 4279 institutions, and published in 403 distinct academic journals, with an average citation count of 18.41 per article.

Quantitative analysis of countries/regions, organizations and author

Analyzing the distribution and collaboration across countries reveals key insights into the global research landscape. The articles analyzed in this study were authored by researchers from 86 countries or regions worldwide, with VOSviewer accounting for articles co-authored by multiple countries. Using VOSviewer, we visualized the global collaboration network in the field of medical innovation and entrepreneurship education, as shown in Figure 2a. Through this figure we can see that there is a strong cooperation between the UK, France, Japan and Poland, while China, Germany, Brazil and South Africa are another close-knit partnership. This depends largely on different economic developments and inter-country communication strategies. Figure 2b presents an overlay visualization generated by VOSviewer, highlighting that the United States, the United Kingdom, and Canada were the early leaders in initiating research in this field, while China, Germany, and Australia have significantly expanded their involvement in recent years. Figure 2c shows the dynamic changes in the annual publication output of the top 15 countries. Since 2015, the United States has seen a sharp rise in annual publications, maintaining a steady output thereafter. Other countries experienced a substantial increase in publication numbers starting in 2019, fueling greater interest in medical innovation and entrepreneurship education globally.

As shown in S. Table 1, most of the top 20 countries by publication volume are developed nations, emphasizing the critical role of economic development in advancing research in medical education. The United States leads with 592 articles, followed by the United Kingdom, Australia, Canada, and

Table 1. Top 20 authors with the most citations.

Rank[a]	Author	Citations[b]	Documents	Average citation per paper	H-Index[c]	Affiliation[d]
1	Frank,Jason R.	1781	5	356.20	34	University of Ottawa
2	Wald,Hedy S.	705	6	117.50	21	Brown University
3	Reis,Shmuel P.	453	4	113.25	23	Holon Inst Technol
4	Scherpbier,Albert J.J.A.	408	5	81.60	49	Maastricht University
5	Borkan,Jeffrey M.	403	4	100.75	21	Brown University
6	Van der Vleuten,Cees P.M.	401	5	80.20	88	European Board Med Assessors
7	Mylopoulos,Maria	383	7	54.71	26	University Health Network Toronto
8	Hendricson,William D.	266	4	66.50	15	University of Texas Health Science Center at San Antonio
9	Woods,Nicole N.	249	5	49.80	26	University Health Network Toronto
10	Teherani,Arianne	231	5	46.20	27	University of California San Francisco
11	Teunissen,Pim W.	225	6	37.50	18	Maastricht University Medical Centre
12	Carney,Patricia A.	148	6	24.67	57	Oregon Health & Science University
13	Santen,Sally A.	135	6	22.50	5	Previously Virginia Commonwealth Univ
14	Ross,Shelley	107	4	26.75	19	University of Alberta
15	Valcke,Martin	88	4	22.00	49	Ghent University
16	Scheele,Fedde	55	4	13.75	30	Onze Lieve Vrouwe Gasthuis OLVG
17	Skochelak,Susan E.	54	4	13.50	16	American Medical Association
18	Chang,Anna	51	4	12.75	7	Johns Hopkins University
19	Liu,Hsing-yuan	45	4	11.25	10	New Taipei Municipal TuCheng Hosp
20	Lomis,Kimberly D.	43	4	10.75	6	Cincinnati Children's Hospital Medical Center

[a]Ranked by article number. Authors with the same number of articles were then ranked by total citations and h-index.
[b]Total citations refer to the sum of citations received these years of each author.
[c]H-Index, is extracted from Web of Science, introduced in detail in the Data Analysis section.
[d]Affiliation for each author is the latest one shown on Web of Science, in order to distinguish different people only.

China. In terms of citation frequency, New Zealand has the highest average citation count, reflecting its significant research quality and influence. However, the United States ranks highest in the h-index, underscoring its dominant academic impact in this field.

Analyzing the publication volume and collaboration patterns of institutions provides valuable insights into their research strength and influence within specific fields [35]. S. Table 2 presents the top 20 institutions by publication volume (with a tie for 14 publications between four institutions – Massachusetts General Hospital, University of Minnesota, University of Ottawa, and University of North Carolina – resulting in 21 institutions in total among the top 20). As shown in S. Table 2, higher education institutions are the dominant contributors to research on medical students' innovation and entrepreneurship education. Among the top 20 institutions, Harvard University (USA), the University of California San Francisco (USA), and the University of Toronto (Canada) lead with 56, 42, and 30 publications, respectively. These universities also have high h-index values, highlighting their substantial impact and productivity in this field. Notably, although the University of Washington (USA) did not rank in the top three for publication volume, its average citation count (42.22) significantly surpasses that of other institutions, reflecting its superior research quality. Furthermore, 15 of the top 20 institutions are based in the United States, illustrating the prominent role of U.S. research institutions in this field.

Using VOSviewer, the visualized collaboration network reveals that the majority of these 43 institutions are located in the United States and Canada. Clusters are primarily centered around Harvard University, the University of California San Francisco (USA), and the University of Toronto (Canada). These clusters exhibit frequent internal communication, while inter-cluster connections remain relatively sparse. The high-frequency collaboration among these three core institutions has played a pivotal role in driving the growth of this research field (S Figure 2).

Further analysis of the authors' contributions, as shown in Table 1, ranks the core authors in the field by citation frequency. This includes details such as their names, publication volume, total citations, average citations per article, h-index, and institutional affiliations. The dedicated efforts and significant contributions of these researchers have greatly advanced medical education. Scholars within the same field often specialize in different research areas. Professor Jason R. Frank from the University of Ottawa is the most cited author in medical innovation and entrepreneurship education over the past two decades, with an average citation count far exceeding that of other authors (total citations: 1781, average citations per article: 356.20). Scholars such as Van der Vleuten, Cees P.M., Carney, Patricia A., Scherpbier, Albert J.J. A., and Valcke, Martin have exceptionally high h-index values—88, 57, 49, and 49, respectively – demonstrating the high quality of their work and widespread academic recognition. Notably, Maria Mylopoulos has published 7 articles, making her the most prolific author in the field. The visualized author collaboration network, created by VOSviewer, shows that collaborations are primarily confined within the same institution, with limited inter-institutional communication (S Figure 3).

These findings together reveal that current innovation and entrepreneurship education for medical student is relatively centralized and lack of extensive international collaboration among various countries and organizations.

Quantitative analysis of journal and article

The 1265 articles we collected were published across 403 distinct journals. According to Bradford's law [29], 7 of these 403 journals are classified as core journals, which demonstrates that these journals have a high level of scholarship and impact (Figure 3a). Figure 3b illustrates the publication trends of the top 15 journals in this research field from 2000 to 2024, with Academic Medicine consistently leading from 2003 to 2021. Notably, BMC Medical Education experienced rapid growth starting in 2019 and has maintained its top position from 2022 to 2024, largely due to a significant increase in its total publication volume after 2019. Figure 3c shows the top 15 journals by publication volume, with BMC Medical Education (138 articles), Academic Medicine (91 articles), and Medical Teacher (70 articles), Anatomical Sciences Education (35 articles), Teaching and Learning in Medicine (34 articles), Medical Education (30 articles) and Medical Education Online (30 articles) occupying the top seven spots. These journals primarily focus on medicine and medical education and are all ranked in the first quartile (Q1) in both the education and healthcare categories of the JCR.

References with high burst intensity tend to cover a wider range of scientific output and have a greater impact on subsequent research [32]. As shown in S. Table 3, the top three references by burst intensity are Medical Student Education in the Time of COVID-19 (burst intensity = 8.05) [36], Perceptions of Medical Students towards Online Teaching during the COVID-19 Pandemic: A National Cross-Sectional Survey of 2721 UK Medical Students (burst intensity = 5.07) [37], and Medical and Surgical Education Challenges and Innovations in the COVID-19 Era: A Systematic Review (burst intensity = 4.96) [38]. These findings clearly illustrate the profound impact of COVID-19 on medical education.

S. Table 4 presents the 15 most-cited articles, which were published in 13 distinct journals. Notably, many of these articles are not featured in the top 15 journals by publication volume in the field of medical innovation and entrepreneurship education. Additionally, several authors are included in the list of high-productivity researchers (Table 1). Some articles focus on the challenges and innovations in medical education due to the pandemic [38–40], while others examine the impact of emerging technologies, such as video games and mobile devices, on medical student education [41–43]. In addition, curriculum reforms and innovations in teaching methods – such as model-based teaching, longitudinal integration, and reflective practice – are key areas of research [44–47]. Furthermore, pain education, nutrition education, and humanities education have also attracted significant attention in medical education [48–50]. Given that these articles are the most cited, it can be concluded that these topics have drawn substantial attention within the field. Notably, many core journals outside the field have been heavily utilized, indicating widespread interest in this area. However, it is important to recognize that citation counts do not perfectly reflect the quality of an article and should not be considered the sole metric of evaluation [11]. In conclusion, interdisciplinary integration remains a vital avenue for future educators aspiring to make a meaningful contribution to the field of medical innovation and entrepreneurship.

Quantitative analysis of keyword co-occurrence cluster and thematic evolution

Cluster analysis of high-frequency keywords enables the identification of core research focuses in the field and provides insights into the internal details of each focus. The following is a manual summary and analysis of 10 clusters (Figure 4a).

In the clustering process, each cluster is associated with specific topics, with smaller cluster numbers indicating a greater number of related keywords. In S. Table 5, 'size' refers to the number of keywords in each cluster, 'Silhouette' measures cohesion (with values closer to 1 indicating higher cohesion), and 'mean (year)' represents the average year of occurrence for each cluster [51]. The last three columns display the results of each cluster under different algorithms.

Using CiteSpace's clustering function, we identified 10 main themes from the research literature, including #0 medical education/undergraduate medical education, #1 clinical competence/augmented reality, #2 radiation oncology/medical education, #3 urology education/medical students, #4 controlled trial/recruitment, #5 students/students, #6 medical students/medical students, #7 clinical skills/clinical education, #8 randomized-comparison trial/flipped classroom, and #9 climate change/planetary health, as shown in S. Table 5. These keywords show a strong alignment with the subsequent analysis of keyword evolution and burst terms, indicating high clustering reliability.

Analysis of the cluster diagram reveals that research has increasingly focused on several key areas within medical innovation and entrepreneurship education.

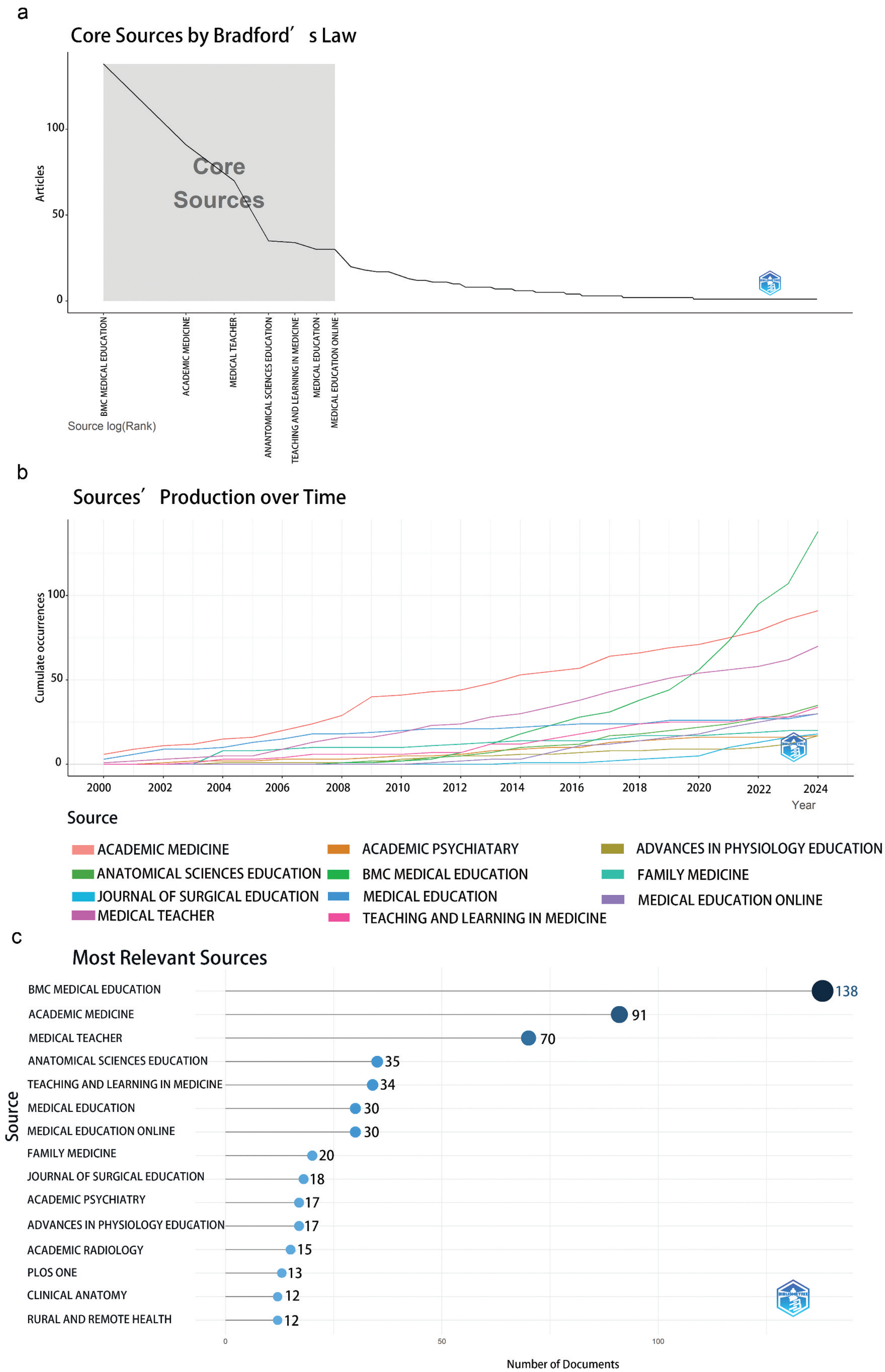

Figure 3. a) to further investigate the distribution of literature in this field, we referred to Bradford's law, proposed by the bibliographer Bradford in 1934. In this law, journals are arranged in decreasing order of the number of articles they publish, and they are classified into three zones: the core zone, the related zone, and the non-related zone. In this distribution model, the number of articles in each zone is equal, and the number of journals follows 1: a: a^2 relationship (where $a > 1$). b) journal output trends within the top 10 from 2000 to 2024. c) top 15 journals in the field of medical students' innovation and entrepreneurship education based on publication counts.

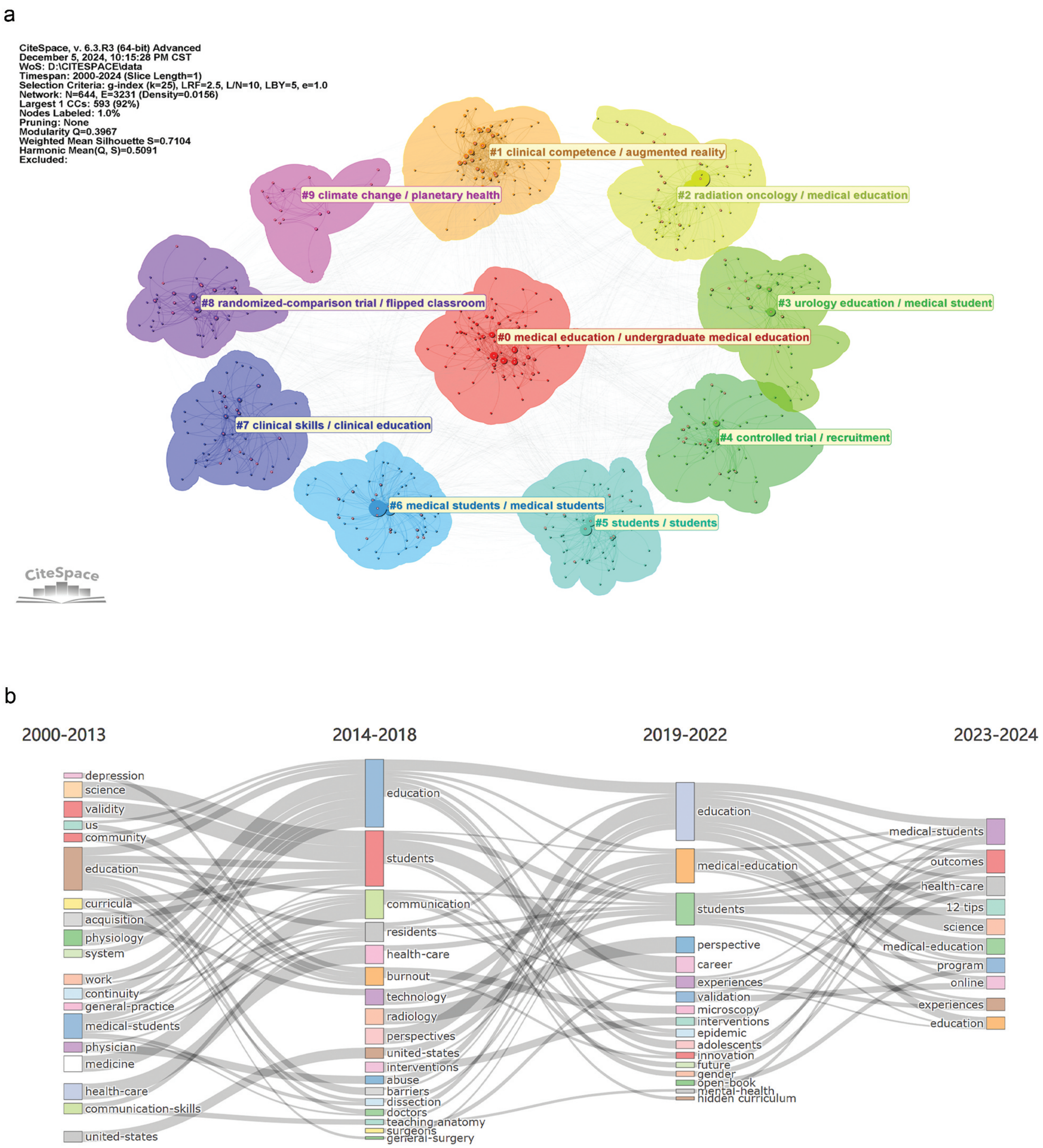

Figure 4. a) a cluster analysis of the keywords from 1,265 articles was conducted using CiteSpace, with the results illustrated in Figure 4a. Each color represents a different cluster, with 10 clusters formed, numbered from #0 to #9. Clusters with smaller numbers contain a larger number of keywords. Each cluster consists of multiple closely related terms, and the lines between nodes indicate the strength of associations among the keywords. b) using the bibliometric package Bibliometricx in the R language environment, the 'thematic evolution' function under the 'conceptual structure' toolbar was employed to visualize the evolution of keywords and establish the dynamic relationships of keyword changes across different time periods, spanning from 2000 to 2024. The time segmentation points were established in 2013,2018 and 2022.

Clinical training and methods (#1 clinical competence/augmented reality, #2 radiation oncology/medical education, #3 urology education/medical student, #7 clinical skills/clinical education)

The importance of clinical skills in medical training is undeniable [52,53]. With advancements in science and technology, new methods such as augmented reality, online learning, distance learning, and virtual models are increasingly applied in clinical training. These approaches offer advantages like convenience and cost-effectiveness, enabling students to complete tasks and access resources more efficiently [54–58]. As a specific example, radiation oncology encompasses findings in molecular and cellular radiation biology, radiation physics, radiation technology, and clinical oncology, which closely integrates medicine with emerging technologies [59]. Notably, it has made significant strides in the interdisciplinary field of medical engineering [60,61], and it also stands as a model for medical-patient communication [62]. Urology education, which has received much attention mainly because of the high incidence of prostate cancer, is also a representative case of the application of new technologies in diagnosis and treatment to specific disciplines [63–65].

Research and teaching methods (#4 controlled trial/recruitment, #8 randomized-comparison trial/flipped classroom)

Controlled trials and randomized-comparison trials are widely adopted in medical education, commonly used for evaluating teaching content, methods, and outcomes [66–68]. For example, as an important part of the healthcare process, patient-centered counseling is increasingly emphasized in the teaching process, and a wide variety of modalities are used to evaluate such as meta-analysis, controlled trial, and other studies [69–71]. Building on these studies, curriculum reform – especially innovations in curriculum structure – has gained momentum [72], including changes in the roles of teachers and students, such as the flipped classroom. Additionally, curriculum design innovations, such as PBL (problem-based learning), have also emerged [73–75]. Notably, artificial intelligence has rapidly integrated into medical education [76,77], with tools like ChatGPT (a generative language model) showing significant potential to transform the field [78,79].

Undergraduate competence and teaching (#5 students/students, #6 medical students/medical students)

Undergraduate medical education is increasingly adopting competency-based education (CBE) [80], focusing on developing competencies and learner-centered approaches to better prepare graduates for clinical practice [81,82]. Given the close link between medical education and clinical practice, simulation has become an essential component of training [83,84]. Apart from simulation, medicine and other disciplines such as art and computing have formed interprofessional and interdisciplinary approaches for undergraduate education [85–87]. Additionally, reflecting on outcomes and the knowledge itself is equally crucial [88], and future innovation and entrepreneurship education should emphasize the encouragement and assessment of reflective practice [89,90].

Global context (#9 climate change/planetary health)

Global health is currently under significant strain from multiple factors, including pandemics, wars, climate change, and resource scarcity [91], with climate change itself posing a serious threat to global health. Consequently, there is an urgent need to educate medical students about climate change and its associated health challenges, one important aspect of which is sustainable healthcare [92,93]. Moreover, the COVID-19 pandemic, which began in 2019, has been both a public health crisis and a governance challenge [94]. It has not only impacted health policies but also disrupted medical students' internships [95,96]. Therefore, educating medical students on planetary health is critical to safeguarding the health and well-being of humanity [97,98].

Cluster analysis reveals that, amid the broader context of climate change, the COVID-19 pandemic, and technological advancements, medical innovation and entrepreneurship education is shifting towards a competency-based, integrated model that combines clinical practice with advanced technologies and innovative educational approaches.

Using the Bibliometricx package in R, we conducted a visual analysis of keyword evolution through the 'Thematic Evolution' function in the 'Conceptual Structure' toolbar, establishing dynamic relationships in keyword evolution across different time periods.

Figure 4b shows the dynamic evolution of keywords over the past 24 years. From 2000 to 2013, the key themes were 'education' and 'medical students', reflecting a focus on education itself and the learners [99]. Between 2014 and 2018, the research themes carried over from the previous period, with 'education' and 'students' remaining prominent, while 'communication' emerged as an important topic, indicating an increasing emphasis on interactivity in medical education [100,101]. 'Technology' means that medical education has begun to focus on co-progressing with scientific developments, an example of which is 'radiology' [102,103]. From 2019 to 2022, 'medical education,' a broad theme, gained widespread academic attention, with 'career' closely linked to 'students,' underscoring the research focus on the impact of the pandemic on medical education and student employment [104–106]. The emergence of the keyword 'epidemic' is strongly associated with COVID-19 [107,108]. In the last two years, research has become more balanced, with 'outcomes' and 'health-care' continuing from prior themes, highlighting the growing body of research on medical education and its shift towards improving broader healthcare competencies [109–112]. 'Online' as an important addition to educational methods should not be ignored after the pandemic [113,114].

Hotsopt evolution analysis

The burst intensity and evolution of keywords offer valuable insights into the hot topics and development trends within the field of medical innovation and entrepreneurship education research. This study employed CiteSpace's Burst Detection function to identify the top 15 keywords by burst intensity, as shown in S. Table 6. As illustrated in the figure, at the start of 2001, 'competence,' 'problem-based learning,' and 'physicians' exhibited the highest burst intensity. 'Competence' of medical students as a key and important topic has maintained a high intensity of outbursts throughout the previous decade, both in terms of competence development and competence assessment [115–118]. PBL, a long-established, student-centered method, has seen widespread adoption [74], likely influenced by surveys in the 1990s in which students and teachers expressed strong satisfaction with its effectiveness [119]. From 2009 to 2018, the burst keywords were 'attitudes,' 'school,' and 'us.' Between 2016 and 2019, 'perceptions,' 'clinical education,' and 'knowledge' emerged as significant research areas, indicating the growing importance of practice-based, patient-centered clinical education in enhancing medical students' knowledge [120,121]. Since 2020, four new research hotspots – 'COVID-19,' 'implementation,' 'virtual reality,' and 'online learning' – have arisen, all of which are closely linked to the pandemic. The implementation of new educational methods became a priority in response to the pandemic's impact [122], with home-based learning and work accelerating the development of online education and VR technologies [123–125]. These profound shifts continue to shape current research, with these keywords remaining central to ongoing studies.

Discussion

Our study provides a comprehensive analysis of global publications on medical student innovation and entrepreneurship education, using data from the Web of Science database (2000–2024). It is the first to employ bibliometric indicators and visual tools to map the development, evolution, and key research directions in this field. Compared to previous studies that have primarily focused on qualitative assessments or several regions, this study employs quantitative methods to highlight the growing interest and the research gaps in this critical area globally, which is becoming increasingly relevant to health education and policy-making [126–128].

The concept of innovation and entrepreneurship education for university students first emerged in the United States during the 1940s and 1950s. Gradually, developed countries such as the United States, the United Kingdom, Japan, and several European nations integrated it into their university systems, refining and expanding its scope [129]. However, progress remained relatively slow. In 2009, the United States incorporated an 'innovation strategy' into its national development blueprint, marking a major milestone. By 2011, this strategy prioritized education, scientific research, and infrastructure as core pillars for national development. In 2015, the strategy further expanded to emphasize investments in science and technology innovation, particularly in manufacturing, precision medicine, and smart cities. For the first time, innovation and entrepreneurship education received national-level advocacy, funding, and support [130]. Similar trends have been observed in other countries, where strategic investments in medical education are increasingly driving the integration of innovation and entrepreneurship into curricula [131]. Our study's results shows a surge in publications after 2015 (S. Figures 1a and 2c), when the concept of precision medicine was just proposed [132], which also coincides with these national policy shifts and underscore the importance of government and institutional support in advancing this area. The surge in publications in 2020 appears linked to the global impact of the COVID-19 pandemic., which forced a reevaluation of traditional educational methods and accelerated the adoption of online learning, creating new opportunities for innovation and entrepreneurship education in medicine as described in the previous section of Thematic Evolution and Hotspot Evolution Analysis. This aligns with similar shifts observed in previous studies [32,33], indicating that major international events such as public health crises often serve as catalysts for educational transformation.

Innovation and entrepreneurship education for medical students began later than in other university disciplines and has encountered significant challenges in implementation. It might greatly result from the specialized nature of medical training – focused on producing healthcare professionals rather than business-oriented individuals – this field has primarily emphasized innovation over entrepreneurship, or addressed innovation alone [133].

As is shown in our results, the development of medical education is characterized by high centralization and low cooperation among different countries. Developed countries, including the United States, the United Kingdom, Australia, Canada, and several European nations, lead in this field through state policies, university-level initiatives, and societal needs (Figure 2, S Figure 2).

Developing countries, particularly China, experienced modest growth in 2020 (Figure 2c), largely driven by transformative changes in teaching models and concepts during the COVID-19 pandemic [134].

Meanwhile, international collaboration is evident in partnerships between nations – especially the United States, the United Kingdom, Australia, and Germany – and their domestic universities (Figure 2a). While these countries serve as pioneers in the field, this dynamic exacerbates regional disparities in innovation and entrepreneurship education for medical students. Consequently, medical talent resources are increasingly concentrated in a few nations, intensifying the polarization of global medical education. This growing inequality demands urgent attention from international educational organizations and institutions to ensure more equitable access to innovative medical education worldwide.

While policy environments provide critical support, economic conditions ultimately play a decisive role in shaping the development of innovation and entrepreneurship education for medical students. The unaffordability of training medical students, who are critically needed by society, remains a key factor influencing the reform of medical education in developing countries. Despite innovative medical education models, the high cost often drives graduates to seek employment abroad as high-income specialists, rather than becoming the innovation-driven or entrepreneurial talents urgently needed in their home countries. Richard Doll, the former UK health minister, aptly stated that 'money is the root of all progress' [135], emphasizing the significant role of financial constraints in the global imbalance of medical students engaged in innovation and entrepreneurship education.

As shown in Table 1 and S. Table 2, higher medical schools remain the primary contributors to innovation and entrepreneurship education, with the top 20 institutions predominantly located in developed countries, led by the United States. A few hospitals, such as the Mayo Clinic, Brigham and Women's Hospital, and Massachusetts General Hospital, also actively contribute to promoting innovation and entrepreneurship education. Quantitative analysis of keyword co-occurrence clusters, theme evolution, and hotspot evolution (Figure 4, S. Table 6) reveals that current research primarily focuses on the learning aspects of medical education, including curriculum content, teaching methodologies, teaching tools, and clinical practice. However, the development of innovation and entrepreneurship abilities among medical students themselves remains significantly underexplored. Innovation and entrepreneurship education aims to cultivate students' abilities in innovative thinking and entrepreneurial practice [136]. For medical students and educators, this involves integrating the development of such skills into teaching and learning activities, forming the foundation of meaningful and impactful educational initiatives. This integration is crucial for producing healthcare professionals capable of driving innovation in practice, as emphasized by recent global reforms in medical curricula [137]. Nevertheless, innovation and entrepreneurship education for medical students faces unique challenges compared to other disciplines. Factors such as memorization-heavy courses, rigid and specialized faculty structures, insufficient integration between basic and clinical medicine, lengthy training periods, and the difficulty of clinically translating medical research have collectively hindered the creation of a comprehensive, in-depth, and evaluable system for innovation and entrepreneurship education in medical schools.

Medical education cannot remain confined within the disciplinary boundaries of medicine alone. With the rapid advancements of the Fourth Scientific and Technological Revolution – marked by developments in virtual reality, robotics, nanotechnology, and artificial intelligence [138,139]—medical education must extend beyond traditional curricula. These technologies have the potential to transform medical training by enabling new learning environments and methods of patient care [140]. It should aim to integrate and expand medical knowledge through updated informational frameworks and the fusion of medicine and engineering, fostering a multidisciplinary medical-industrial crossover. By instilling innovative thinking, entrepreneurial qualities, and practical skills during undergraduate training, this approach has the potential to prepare students to meet the evolving demands of healthcare and propel medicine into a new era of transformative development. As shown in Figure 4 and S. Table 6, global medical education is increasingly moving toward this interdisciplinary direction.

As shown in the analysis of references and journals from highly cited publications and high-citation outbreaks (S. Table 3, S. Table 4, Figure 3b), the global COVID-19 pandemic has profoundly influenced medical student education, reshaping the developmental trajectory of medical innovation and entrepreneurship education. This highlights the urgent need for a robust framework to address extreme and unpredictable conditions. Such a system is essential for cultivating the innovative spirit and entrepreneurial capabilities of medical students, enhancing their competitiveness and adaptability in a rapidly evolving world [133]. Despite disruptions caused by challenges such as epidemics, wars, climate change, and resource shortages, history demonstrates that major advancements in science and technology often arise from crises [141]. For medical students, preserving human health must extend beyond disease treatment to emphasize prevention and health promotion, aligning with a broader 'one-health' perspective [142]. With medical universities at the core, integrating resources from organizations, institutions, and enterprises is crucial. Collaborative efforts – such as partnerships, lectures, research initiatives, and practical applications of innovations – aim to establish an organic, dynamic, and interactive health eco-education system.

These findings have far-reaching implications for healthcare policy and underscore the urgency of advancing curricular reform in global health. What's more, the research contributes to aligning medical education with the needs of modern healthcare systems and to the global realization of universal healthcare coverage (UHC) proposed by WHO [143].

Conclusion

In conclusion, this paper uses bibliometric and visualization methods to systematically map the global research landscape of medical student innovation and entrepreneurship education since 2000. The results of our study reflect a growing awareness of the importance of early introduction of innovation and entrepreneurship, due to technological advances and changing healthcare needs. This study also highlights the integration of medicine with other disciplines such as engineering, technology and business as a major trend. Practically, it provides feasible recommendations for medical schools, policy makers and healthcare organizations. Collaborative efforts by all stakeholders will make the training of healthcare professionals more effective in stimulating medical innovation and improving global health. Finally, the study lays the foundations for the development of future medical education policies.

Recommendations

Based on the above discussion, we've offered several key recommendations to improve the current situation. In terms of curriculum, medical schools should integrate the development of innovative and entrepreneurial skills into the compulsory curriculum and encourage students to actively carry out interdisciplinary programs aiming to cultivate their cross-field as well as multi-dimension thinking skills. After students have acquired these basic abilities, schools can consider cooperating with relevant enterprises or research institutes so as to provide a platform for students to realize their creative thinking, which will help them better solve the problems in the real world. In order to ensure the integrity of the education chain, the effectiveness of innovation and entrepreneurship education should be continuously paid attention to in the future, and the development of a complete and scientific evaluation system will help optimize the curriculum and prepare outstanding medical students to lead innovation and entrepreneurship in the field.

Limitations

There are various restrictions on this study. First off, we only used the Web of Science database, which may have led to the exclusion of relevant articles from other databases in this field from our study. (e.g., Scopus, PubMed). Second, the analysis only considered articles and reviews, leaving out other publication types such as editorials, letters, and conference abstracts. It could have reduced the comprehensiveness of our research. Furthermore, since our study focused exclusively on English-language publications, the results may be subject to linguistic and geographic bias. At last, there are intrinsic methodological limitations in bibliometric analyses, therefore algorithmic classifications in bibliometric tools are subjective by nature based on preset parameters for data extraction and processing. Despite these limitations, our study provides a profound insight into innovation and entrepreneurship education of medical students, establishing a framework for future investigations in this domain.

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Author contributions

TW designed the study. XYW and FZ performed the search. TW, XYW, FZ and XRZ analyzed the data and wrote the manuscript. All authors contributed to the article and approved the submitted version.

Data availability statement

Datasets generated for this study will be made available by the authors, to any qualified researcher upon request.

Ethical approval

Not applicable. This study does not involve human subjects.

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